



HORMONES, SEXUAL HEALTH & MISC

B12 (Methylcobalamin)

B12 – 10 ml per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	Methylcobalamin (vitamin B12 active coenzyme form)
SYNONYMS / ALIASES	MeCbl; mecobalamin; methyl-B12; coenzyme M; co-methylcobalamin
CAS NUMBER	13422-55-4
MOLECULAR FORMULA	C63H91CoN13O14P
MOLECULAR WEIGHT	1344.40 g/mol
AMINO ACID SEQUENCE	Not applicable - vitamin / cobalt-corrinoid coenzyme, not a peptide
SOURCE	Semi-synthetic (bacterial fermentation followed by chemical conversion to the methyl form)

PHYSICAL PROPERTIES

APPEARANCE	Dark red to red-violet aqueous solution (color characteristic of the corrinoid Co(III) chromophore)
PURITY SPECIFICATION	>=99% by HPLC (confirm against batch COA)
CONTAINER	10 ml aqueous solution per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	2-8 C, protect from light
SOLUTION STABILITY	Methylcobalamin is photosensitive: ambient light converts it to hydroxocobalamin. Store in amber glass or wrapped in foil.
SUGGESTED DILUENT	Not applicable - supplied as ready solution
SHELF LIFE (SEALED)	Typically 24 months at 2-8 C, protected from light
STABILITY NOTES	Highly photosensitive. Avoid prolonged exposure to fluorescent or daylight. The Co-C bond is photolabile.

STRUCTURAL NOTES

Methylcobalamin is one of the biologically active coenzyme forms of vitamin B12 (cobalamin). Structurally, it consists of a corrin macrocyclic ring (related to porphyrin but with a direct C-C bond between two of the pyrrole-like rings) coordinating a central cobalt(III) ion, with a methyl group as the upper axial ligand on cobalt and a 5,6-dimethylbenzimidazole nucleotide as the lower axial ligand. The cobalt-carbon bond linking the methyl group to the corrin core is the defining structural feature of methylcobalamin and is one of the few naturally occurring organometallic bonds in biology.

SELECTED PEER-REVIEWED REFERENCES

1. Banerjee R., Ragsdale S.W. - *The many faces of vitamin B12: catalysis by cobalamin-dependent enzymes* - Annual Review of Biochemistry, 2003 - DOI: 10.1146/annurev.biochem.72.121801.161828
2. Watanabe F. - *Vitamin B12 sources and bioavailability* - Experimental Biology and Medicine, 2007 - DOI: 10.3181/0703-MR-67
3. Hodgkin D.C. et al. - *Structure of vitamin B12* - Nature, 1956 - DOI: 10.1038/178064a0

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

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Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.